

Graph based framework and its implemented prototype for personalized study planning

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Abstract — The fast technological development dictates the growing demand for knowledge and skills. Thus, it is necessary to offer learners a more flexible way of knowledge acquisition, in which they can plan their studies independently. The studies are adjusted individually to the needs of each learner, because not everybody has the opportunity to participate in traditional learning process. The authors offer a graph-based framework consisting of four graphs to plan and carry out each individual study process. Within the framework every learner can design his/her own study programme, which is described by AND – OR graph. The second graph represents the structure of a study course by showing relationship between topics. The third graph is constructed by mapping each topic to the graph, which includes all topics and their relationships. The fourth graph represents a topic and its relationships with all needed learning objects to acquire knowledge about the topic. The proposed framework has been implemented as a personalized study planning system prototype SPS, which is also described in this article.

Keywords - intelligent tutoring system, tutoring module, personalized education, graphs.

I. INTRODUCTION

Nowadays not only the significance of education for different groups of population is discussed more often, but also the accessibility assurance based on different needs. The planning documentation of Latvia, European Commission, and Finland stress the accessibility of education. To promote it further on, the emphasis relies on:

- e-school [1];
- use of information and communication technologies (ICT) in teaching [1, 2, 3, 4, 5];
- lifelong learning [1];
- diversity of studying modes [2];
- use of ICTs in drafting the learning environment [4, 5].

One can obtain education in learning process using different forms of training. There are several typical tutoring forms such as full-time, part-time or evening, external, and distance learning. The most demanded form of studies is full-time studies [6]. Taking into account the dynamic rhythm of life, educational institutions should offer other flexible tutoring

forms in order to improve the level of accessibility of education for different groups of society. The presence of ICTs initiated the creation of technology-supported learning, which brought new opportunities into the tutoring process. Technology-based learning uses ICTs to support learning and teaching, and lifelong development of competence [7]. Technology-based learning includes the use of such technologies as presentation programmes, video conferences, webpages, e-learning and m-learning [8]. It gives essential advantages for assuring the tutoring accessibility and the interactivity. However, one cannot forget the decisive significance of the teacher in the organization and management of the learning process. In order to make the learning process more flexible, personalized, attractive and interesting for the learner, at the same time not losing the effect of presence of the teacher during the learning process, the solution is Intelligent Tutoring Systems (ITS). Nowadays ITSs are analysed very intensively in literature, for example in [9, 10, 11, 12]. ITSs are computerized tutoring systems, supplemented with artificial intelligence conceptual aspects. Originally ITS contained three components: a domain, a teacher (instructor) and a student module. Later on one more component - the interface module was added. The description of modules in more details is in the next chapter. The advantage of ITSs compared to computerised tutoring systems is the personalization of tutoring. Personalization as an important factor in ITS has been analysed by many researchers [13, 14, 15]. It foresees the assurance of personalized tutoring for each learner taking into account his/her desires and characteristics of learning. Therefore, the authors of this paper propose a graph-based framework for tutoring module of ITS for individual study planning, which has been implemented in personalized study planning system prototype SPS. This framework allows each student to tailor a study programme by adapting the modularized curriculum structure and to choose the suitable learning strategy for each study course.

The next section of this paper shortly outlines the structure of intelligent tutoring system. The third section presents existing personalized learning construction systems. The fourth section describes the architecture of personalized study planning system. The fifth section is devoted to personalized study planning systems SPS prototype. The sixth section describes SPS integration with intelligent knowledge assessment system (IKAS) based on concept maps and

Moodle. The conclusion gives a short summary and outlines directions for future research.

II. THE STRUCTURE OF INTELLIGENT TUTORING SYSTEM

Analyzing literature sources [16, 17, 18] one may conclude, that, basically, intelligent tutoring systems consist of the domain module, the student module, the teacher or the instructor module and the interface module. It should be noted, that the basic components in different literature sources [19] may be named differently but the essence is the same.

The modules of intelligent tutoring system have the following functionality [16, 18]:

- the tutoring (teacher) module functions as a helper in the tutoring process, adjusts the tutoring process to the learner's knowledge, capability, capacity and will; and stores knowledge about tutoring methodology;
- the student module assesses learner's psychology, mentality, and pre-training skills and knowledge level, stores evaluation results, that helps to select most appropriate study material during the tutoring process;
- the expert module contains the knowledge on what to teach. It includes the set of rules, which help to solve any problem of learner and teacher.

From aforementioned, it is obvious, that personalized study planning is an essential component of the tutoring module. Therefore, the proposed framework is designed in a way, that allows integrate it with some ITSs at least (see Section 6).

Next chapter is devoted to the existing personalized learning construction systems.

III. PERSONALIZED LEARNING CONSTRUCTION SYSTEMS

As the first step of personalized study planning is the creation of the study programme, this chapter is devoted to the studying programme construction tools. The most often mentioned tools for personalized studying programmes are described below.

Kabicher and Motschnig-Pitrik [20] have elaborated the tool for studying programme visualization using the CEWebS (Cooperative Environment Web Services) platform, which is designed to visual description of studying programme using the graph. Description of the studying programme in a form of a graph allows to perceive the total structure of the studying programme, showing the modules, courses and their mutual relationship. Such way of description makes the studying programme more "transparent". It allows to manipulate the programme more easily and, as a consequence, to make it more flexible for the learner. However, it is meant only for teachers; the learners cannot work with it independently. Besides, the studying programme showed in the system is static not dynamic, i.e., all learners can have only one tutoring scenario.

Gestwicki [21] and later Toombs who joined him [22] elaborated the tool for visualisation of the studying programme CurricVis. The tool automatically generates the studying programme as a directed graph, based on information presented in the knowledge basis. CurricVis provides the

design of the programme, that presents its structure and, in addition, the progress view. It is the view of learner's progress of the studying programme acquisition. Compared to the previous tool, CurricVis allows to be operated not only by the designers of the study programme but also by learners themselves, thus enabling them to participate in the studying programme planning process. CurricVis tool does not have an interactive graph creation mode. It means that the learner in the basic study programme can only choose acquirable study courses in the progressive view offered by the programme. One can follow the progress of the programme acquisition, but the learner does not have the created graph of the chosen study courses of the studying programme. The offered programme, like the previously described one, does not provide further tutoring processes, such as training, knowledge assessment, etc.

Zucker in [23] offers the tool ViCurriAS that includes two modules. The study programme module is foreseen for the design and modification of the study programme. The module allows the faculty methodologists and consultants arrange the courses in the study programme and define the pre-requisitions of these courses. The Consultation module allows the consultants to enter the marks or planning information in the same graph, created in the Study programme module. The Consultation module allows to import the study results of the learner's, who has been studying for half a year or a year in another university study programme, into the ViCurriAS system. ViCurriAS system, just like the two previously described systems, does not offer the learner to create the study programme dynamically him/herself, neither realizes further training processes.

Auvinen [24] offers a different approach from the previously described tools. This system describes relationships not between the study courses but between learning outcomes. The system offers the design of study programme using graphs of learning outcomes. The construction of the study programme is based on choice of competences. Then, according to the chosen competences, the graph of acquirable study courses is created, which is related to learning outcomes. Compared to all previously described systems of creation of the study programmes, the learner is able to design his/her own study programme. Unfortunately, such system could be very difficult to implement in Latvia, because then it is necessary to review the definition of the learning outcomes and competences. Additionally, the fact, that the learner may ignore topics that are related to learning outcomes, which are not related to his/her competences, should be regarded as a negative trend.

Nkambou, Gauthier and Frasson [25, 26, 27, 28] offer the tool CREAM (*Curriculum REpresentation and Acquisition Model*) for designing ITS curricula and courses. In the CREAM curricula is represented by three models: the domain, the pedagogy and the didactic. The domain model using capability model represents and organizes domain knowledge. The pedagogy model describes instructional objectives. The didactic model defines learning resources that are necessary to acquire the capabilities and the instructional objectives. The CREAM tool allows to design the curriculum according to a method, that is integrated into CREAM tool and generating courses from the curriculum.

Analysis of literature [22-30] allows us to conclude, that in order to cover all personalized study planning completely, there is a need for two more graphs to describe concept maps and learning objects besides the study programme and course graphs. In the next chapter the authors provide the graph-based framework for personalization of education process consisting of four graphs.

IV. ARCHITECTURE FOR PERSONALIZED STUDY PLANNING SYSTEM

The authors of this paper in the previous work had analysed knowledge representation forms and described the architecture of learning system using four graphs [29]. The graph-based framework for personalization of education process consists of four graphs constructed sequentially [30]:

- a graph $G_1(V_1, Q_1)$, representing a conceptual structure of study programme, allows to design individual study programme;
- a graph $G_2(V_2, Q_2)$, representing a study course, allows to develop individual learning scenario;
- a graph $G_3(V_3, Q_3)$, visualizing each topic using concept map, ensures mapping of each topic to the corresponding concept map;
- a graph $G_4(V_4, Q_4)$, representing learning objects, describes each concept through learning objects.

The authors focus on the scenarios, where learners start personalization by selecting the sequence of study courses to build their individual study plans [30]. This first personalization phase, supported by the framework, is based on AND-OR graph $G_1(V_1, Q_1)$, where V_1 is a set of nodes, which in this case represent hierarchy of study programme, Q_1 represents relationships among nodes. The root node represents the name of the study programme and is labelled by the total number of ECTS (European Credit Transfer System) that is needed for graduation. Descendants (child nodes) of the root node represent study years with corresponding ECTS, while the third level nodes correspond to terms of study years with appropriate ECTS [30]. The nodes at the fourth level represent traditional modularized curriculum structure, namely, compulsory (mandatory) courses, restricted electives and free-choice courses [30]. Predecessor nodes between all mentioned levels are connected with their descendants through AND logical operation. Nodes on the fourth and fifth levels (that represent particular courses) can be connected through logical AND or OR operations. That is because in case of compulsory courses all of them have to be taken into consideration; while in case of restricted electives and free-choice courses only limited numbers of them (defined by the programme) are required [30].

The usage of personalized learning process in this case is quite limited, because each student has alternatives only to select restricted electives and free-choice courses [30]. The graph $G_1(V_1, Q_1)$ clearly shows the content of the study programme and the sequence of courses taught term after the term. The graph $G_1(V_1, Q_1)$ may be transformed from the tree to the network structure by adding cross-links representing prerequisites for courses [30]. The graph includes independent

nodes, which are unlimited by sequence, and which are independent from the previous and following ones. The dependent nodes are restricted by sequence, which describes the subjects order. Students can personalize their studies by choosing sequences of courses from restrictive electives and the set of free-choice courses. Finally, if regulations of university allow a more flexible construction of individual study plan, the graph can fully support personalization [30]. Fig. 1 shows the overall structure of this graph.

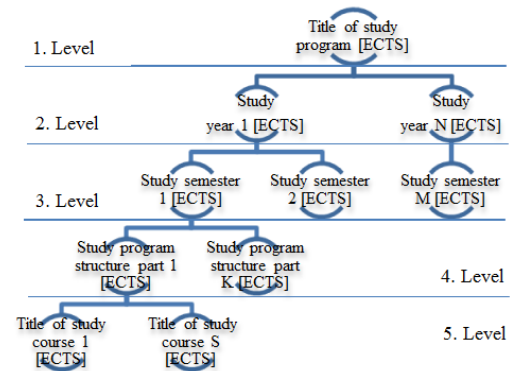


Fig. 1. Part of conceptual structure of the study programme [30]

The next step of the framework is construction of the graph $G_2(V_2, Q_2)$ for each study course. The root node represents the name of the course, and its descendants represent topics with prerequisite relationships between topics (nodes). The framework suggests also a modification of the graph, where the third level nodes are added, representing the structure of each topic, namely, the way of its teaching - lectures, practical seminars, laboratory works and individual works. An illustration of the graph is depicted in 0

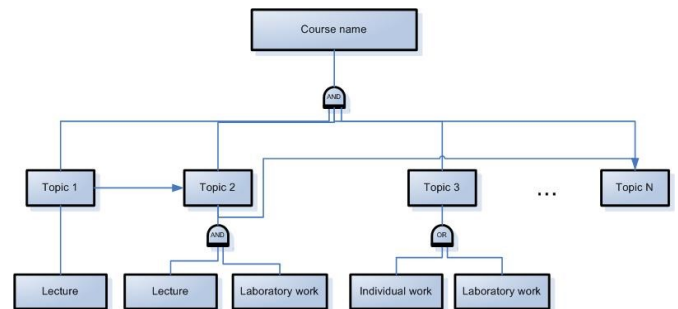


Fig. 2. The structure of study course

If the graph $G_2(V_2, Q_2)$ contains logical operation OR then students have alternative ways of acquiring knowledge of this topics (in Fig. 2, for example, the topic 3 may be taught in form of laboratory or individual work, i.e., students can personalize their learning scenario). Besides, the nodes of this graph may have labels (not shown in Fig. 2), which indicate at what level of Blooms Taxonomy the topic should be learned. Therefore, each student can personalize his/her level of knowledge of particular topic depending on his/her specialization and interests [31].

The third step proposed in the framework is mapping of each topic to the corresponding concept map (CM). CM is a semi-formal knowledge representation tool visualized as a

graph [32]. This graph $G_3(V_3, Q_3)$ consists of a finite, non-empty set of nodes V_3 representing concepts and a finite, non-empty set of arcs Q_3 representing relationships among concepts. Graphs visualizing CM's themselves may be undirected or directed, homogenous or heterogeneous, and also may be represented as attribute graphs (arcs have linking phrases specifying the kind of relationships between a concept pair). The variety of graph is described in [33].

The graph $G_3(V_3, Q_3)$ can be extended by connecting each concept with nodes representing the structure of each topic (see Fig. 2). Thus, students can personalize the way, how they acquire knowledge about a particular concept.

The forth and the last graph is $G_4(V_4, Q_4)$. The root node of this graph represents a concept and the arcs represent learning objects, which are available to acquire the knowledge about it.

According to the standards, for example, SCORM [34] a learning object is an entity, that contains information on study content. From literature [34, 35] it may be concluded, that learning objects can be digital or non-digital, they can be used with educational technologies once or several times, they may be presented in different forms (as a text, video, audio, etc.), used in different courses, may be modular, inter-connectable, dividable, combined in different ways and easy integrated in different courses. Mentioned characteristics of learning objects give clear picture about their role in personalized learning. At the same time it is needed to point out, that the effectiveness of personalization depends very much on the quality of student model implemented in the ITS (due to the scope of this paper this aspect is not discussed here). In addition, from the learning object usage point of view, there are two sequences of learning objects, which may be used for knowledge assessment or self-assessment [36].

More details on the graph representing the study course, the graph visualizing each topic of the used concept map and the graph representing learning objects may be found in [29, 30].

V. PERSONALIZED STUDY PLANNING SYSTEM'S SPS PROTOTYPE

The previous section describes the concept of graph-based framework for personalization of education process while this section is devoted to the developed personalized study planning system's SPS prototype.

Personalized study planning systems prototype SPS provides the following opportunities:

- design of individual study programme (development of individual study programme, conformation of a study programme to the student's mobility results, design of individual training plan within lifelong learning courses, initiation of studies later, e.g. existing study courses, elaboration of individual study plan after knowledge and skills certification and credit point allocation.);
- design of study course structure;
- design of study course structure from topics;
- description of each concept through learning objects.

In the SPS prototype system data base (*SPSDB*), data view and insert forms (*SPS.DataManager*) and study programme construction software application (*SPS.StudentStudyProgramBuilder*) is implemented. *SPS* prototype has been elaborated using *Microsoft.NET Framework 4.5* software creation platform [37, 38]. The following built-in technologies are used:

- *Windows Presentation Foundation (WPF)* [39] – to create graphical user interface;
- *LightSwitch* [40, 41] – to create data review and types of input.

As the system of database management for the local database the *SQL Server 2012 Express LocalDB* [42] is used. It simplifies the installation of the prototype, because there is no need to install and configure the database server. As the *SPS* does not use specific opportunities of *SQL Server 2012 Express LocalDB* the solution can be used also for the other versions of *SQL Server* [43].

The prototype has been elaborated using the software creation tool *Visual Studio 2012* [44]. Implementing the operations with graphs and graph visualizations, open code software libraries *QuickGraph* [45] and *Graph#* [46] are used. In Fig. 3, one can see the components of the prototype.

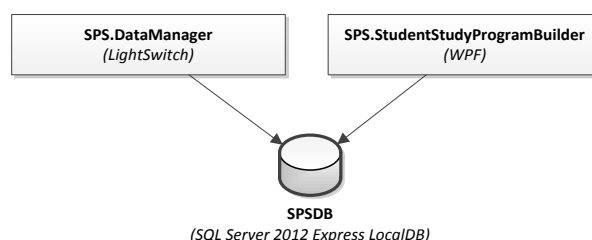


Fig. 3. SPS architecture

Personalized study planning system's SPS prototype contains two tools:

- *SPS.DataManager*, which is designed to input data on study course field, study programme, parts of the study programme, teachers, courses, the structure of each course topics, course topics and students, and student study programmes in order to fulfil personalized tutoring (shown in Fig. 4);

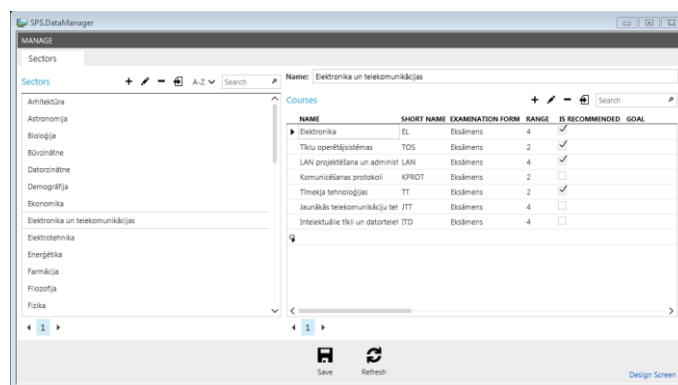


Fig. 4. SPS DataManager

- *SPS.StudentStudyprogramBuilder*, which is designed to determine personalized study programme (see Fig. 5) and the sequence of acquisition of study course topics (see Fig. 60).

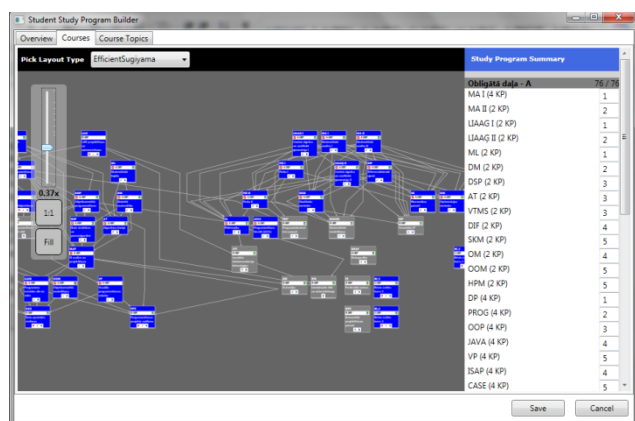


Fig. 5. SPS.StudentStudyprogramBuilder personalized study programme

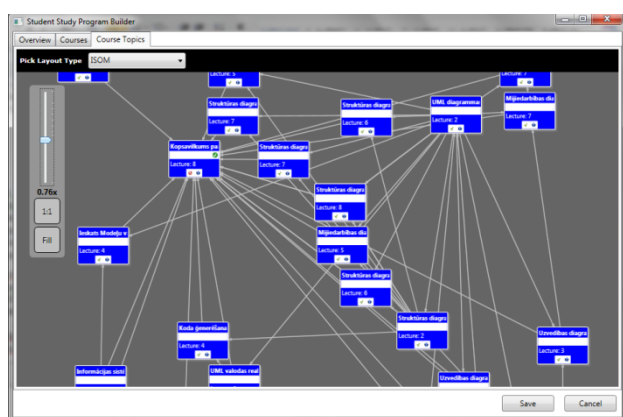


Fig. 6. SPS.StudentStudyprogramBuilder personalized study course acquire

For the personalized study planning system four groups of users are distinguished: learners; study programme directors, teachers and study programme administrators.

VI. SPS INTEGRATION WITH OTHER SYSTEMS

The elaborated prototype allows the learner to create his/her own individual study programme and the sequence of study course topics. To test the knowledge acquired by the learner the system can be integrated with other tutoring systems. This section very shortly outlines two systems with which the elaborated system could be integrated.

The first is the intelligent knowledge assessment system (IKAS) based on concept maps. Since 2005 the IKAS has been elaborated [47] and improved [48] in the Department of System Theories and Design at Riga Technical University. The IKAS is the knowledge assessment system that evaluates learner's knowledge in the particular study course comparing the concept map created by the teacher and the map constructed by the learner.

The second system for possible integration of the prototype was chosen Moodle because it is often used in the universities of Latvia and also in some other educational centres. Moodle

(*Modular Object-Oriented Dynamic Learning Environment*) is an open code study course management system or virtual study environment. Moodle is popular not only in Latvia but also all over the world. This study environment is used in 232 countries by 79 438 organizations [49].

VII. CONCLUSIONS

In this article the authors describe the approach, that allows to offer personalized education process through the graph-based framework and its implementation. The elaborated graphs are integrated into a united database architecture. The SPS prototype of personalized study planning, which allows the learner to design the study programme has been developed, based on the described graph-based framework. It offers to get a visual insight about study course or mutual relationships of the topics. Despite the fact, that other authors also have developed personalized tutoring systems, which are mentioned in the third section, none of them offers the learner personalized study process. After designing the personalized study programme the learner may continue the training study course topics in the graph, which is related with the study programme graph. In the graph of study course topics the learner may choose the sequence of study course acquisition by choosing the topics to acquire. Such system describes graphically the structure of the study programme and study courses, in creation of which the learner may take active part. Thus the learner acquires more clear view on mutual relationships between the study courses and topics of each study course.

It is worth to add that the usage of the graphs as the basic structures in the personalized study planning allows to carry out the analysis of the programme and/or course quality of the structure by doing ranking and identifying courses and topics, which play important role in the study programme. This factor should be taken into account during modification of the study programme or study topics. Due to the scope of this paper, the abovementioned opportunities are not described.

The proposed system's SPS prototype is unique because the analysis of available literature shows that the systems with similar functionality do not exist at all. Higher education reform launched recently in Latvia outlines relevance of personalized study planning.

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